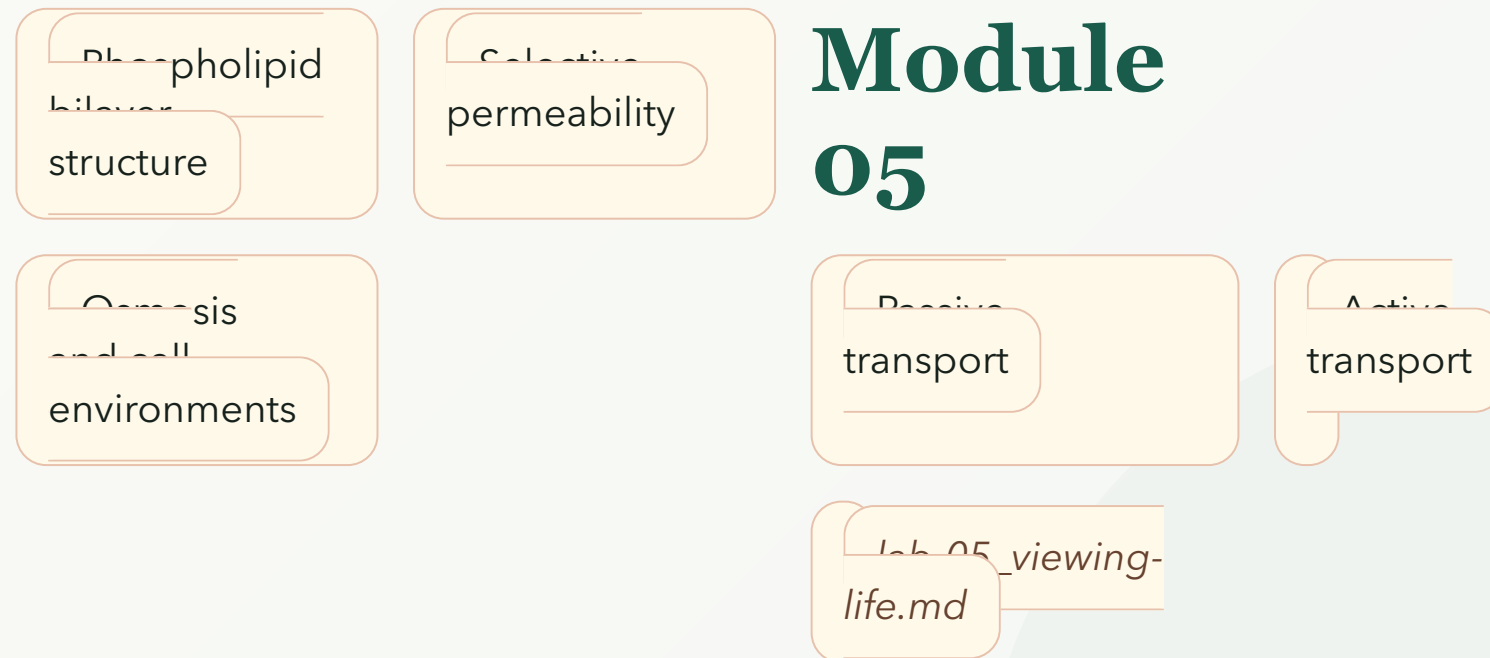


MEMBRANES

Module map

- Module 05: Membranes
- Connected lab: lab-05_viewing-life.md
- Use this deck as the visual path through the module's evidence and retrieval work.



MEMBRANES

Learning objectives

- Describe how phospholipid structure creates a bilayer.
- Predict which molecules cross membranes easily.
- Compare passive and active transport.
- Use concentration gradients to explain movement across membranes.
- Predict osmosis outcomes in hypotonic, isotonic, and hypertonic conditions.

OBJECTIVE 1

Describe how phospholipid structure creates a bilayer.

OBJECTIVE 2

Predict which molecules cross membranes easily.

OBJECTIVE 3

Compare passive and active transport.

OBJECTIVE 4

Use concentration gradients to explain movement across membranes.

OBJECTIVE 5

Predict osmosis outcomes in hypotonic, isotonic, and hypertonic conditions.

MEMBRANES

Topic sequence

- Phospholipid bilayer structure: The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.
- Selective permeability: Selective permeability controls which materials cross easily and which require help.
- Passive transport: Diffusion, osmosis, and facilitated diffusion move substances down gradients.
- Active transport: Active transport uses energy to move substances against gradients.
- Osmosis and cell environments: Tonicity predicts water movement and cell volume changes.

TOPIC 1

Phospholipid bilayer structure

TOPIC 2

Selective permeability

TOPIC 3

Passive transport

TOPIC 4

Active transport

TOPIC 5

Osmosis and cell environments

MEMBRANES

Module 05: Membranes Evidence Map

- Module focus: Membranes
- Central claim: Membranes explains membrane structure with gradients, transport, and cell survival predictions.
- Key nodes to connect: Phospholipid bilayer structure, Selective permeability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-05_viewing-life.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

MEMBRANES

Module 05: Membranes Reasoning Sequence

- Module focus: Membranes
- Trace the model: Phospholipid bilayer structure -> Selective permeability -> Passive transport -> Active transport
- Inputs become outputs: Phospholipid bilayer structure, Selective permeability -> Describe how phospholipid structure creates a bilayer., Predict which molecules cross membranes easily.
- Feedback check: lab-05_viewing-life.md evidence checks whether students can explain passive transport.
- Lab connection: lab-05_viewing-life.md gives students a process to observe or test.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

MEMBRANES

Terms as evidence handles

- Phospholipid: Molecule with a hydrophilic head and hydrophobic tails.
- Bilayer: Double layer of phospholipids forming a membrane.
- Selective permeability: Property of allowing some substances through more easily than others.
- Diffusion: Net movement from high to low concentration.
- Osmosis: Diffusion of water across a selectively permeable membrane.
- Facilitated diffusion: Passive transport through membrane proteins.

PHOSPHOLIPID

Molecule with a hydrophilic head and hydrophobic tails.

BILAYER

Double layer of phospholipids forming a membrane.

SELECTIVE PERMEABILITY

Property of allowing some substances through more easily than others.

DIFFUSION

Net movement from high to low concentration.

OSMOSIS

Diffusion of water across a selectively permeable membrane.

FACILITATED DIFFUSION

Passive transport through membrane proteins.

MEMBRANES

Lab connection

- Lab file: lab-05_viewing-life.md
- Lab evidence should test or illustrate: Passive transport
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Phospholipid bilayer structure

EVIDENCE

lab-05_viewing-life.md

CLAIM

Describe how phospholipid structure creates a bilayer.

MEMBRANES

Contrast check

- Do not stop at: The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.
- Stronger explanation: Tonicity predicts water movement and cell volume changes.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.

DEEPER

Tonicity predicts water movement and cell volume changes.

MEMBRANES

Module 05: Membranes Retrieval and Lab Check

- Module focus: Membranes
- Retrieval prompt: Why do phospholipids form bilayers in water?
- Answer check: Describe how phospholipid structure creates a bilayer.
- Required terms: Phospholipid, Bilayer, Selective permeability
- Lab connection: lab-05_viewing-life.md; lab-05_viewing-life.md supplies evidence for viewing life through structure, boundary, and environment evidence.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

MEMBRANES

Practice quiz bridge

- 1. Which statement best defines Phospholipid in Module 05?
- 2. Which learning goal best supports the topic "Selective permeability"?
- 3. How should lab-05_viewing-life.md support Module 05?
- 4. Which retrieval move best prepares a student for Membranes?

QUIZ 1

A

QUIZ 2

B

QUIZ 3

C

QUIZ 4

D

MEMBRANES

Synthesis and exit ticket

- Exit claim: explain phospholipid bilayer structure using evidence from lab-05_viewing-life.md.
- Must include: Phospholipid, Bilayer, Selective permeability.
- Revision prompt: what evidence would change your explanation?

CLAIM

Phospholipid bilayer structure

EVIDENCE

lab-05_viewing-life.md

REVISION

What would change your mind?